



# Evaluating Differential Equations

Calculus Worksheet · Grade 11-12

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## Learning Objectives

- Find general solutions to first-order differential equations by integration
- Determine particular solutions using initial conditions
- Apply integration techniques to solve separable differential equations

Solve each differential equation by integrating both sides, and find the particular solution when an initial condition is given.

### 1. Find the general solution to the differential equation.

$$\frac{dy}{dx} = e^x - 6x^2$$

Answer: \_\_\_\_\_

### 2. Find the particular solution that passes through the point (1, 0).

$$\frac{dy}{dx} = e^x - 6x^2$$

Answer: \_\_\_\_\_

### 3. Find the general solution to the differential equation.

$$\frac{dy}{dx} = 4x^3 + 2x$$

Answer: \_\_\_\_\_

### 4. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \sin(x) + \cos(x)$$

Answer: \_\_\_\_\_

### 5. Find the particular solution that passes through the point (0, 1).

$$\frac{dy}{dx} = 3x^2 - 4x$$

Answer: \_\_\_\_\_

### 6. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \frac{1}{x} + e^{2x}$$

Answer: \_\_\_\_\_



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7. Find the particular solution that passes through the point (1, 2).

$$\frac{dy}{dx} = 6x^2 + 2$$

Answer: \_\_\_\_\_

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8. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \sqrt{x} + x^2$$

Answer: \_\_\_\_\_

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9. Find the particular solution that passes through the point (0, 3).

$$\frac{dy}{dx} = e^x + 2x$$

Answer: \_\_\_\_\_

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10. Find the general solution to the differential equation.

$$\frac{dy}{dx} = 5x^4 - 3x^2 + 7$$

Answer: \_\_\_\_\_





Remind students to include the constant of integration  $C$  for general solutions and to substitute initial conditions to find  $C$  for particular solutions.

## Solutions

1. Find the general solution to the differential equation.

$$\frac{dy}{dx} = e^x - 6x^2$$

- Cross-multiply to separate variables:  $dy$  equals the right side times  $dx$
- Integrate both sides of the equation
- The integral of  $dy$  is  $y$ , and integrate each term on the right side
- Simplify the coefficient six divided by three to get two
- Add the constant of integration  $C$

**Answer:**  $y = e^x - 2x^3 + C$

2. Find the particular solution that passes through the point  $(1, 0)$ .

$$\frac{dy}{dx} = e^x - 6x^2$$

- Find the general solution by integrating both sides
- The general solution is  $y$  equals  $e$  to the  $x$  minus  $2x$  cubed plus  $C$
- Substitute  $x$  equals 1 and  $y$  equals 0 into the equation
- Solve 0 equals  $e$  minus 2 plus  $C$  for  $C$
- Find  $C$  equals negative  $e$  plus 2 and write the particular solution

**Answer:**  $y = e^x - 2x^3 - e + 2$

3. Find the general solution to the differential equation.

$$\frac{dy}{dx} = 4x^3 + 2x$$

- Separate variables by multiplying both sides by  $dx$
- Integrate both sides of the equation
- Apply the power rule to each term
- Add the constant of integration  $C$

**Answer:**  $y = x^4 + x^2 + C$

4. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \sin(x) + \cos(x)$$

- Cross-multiply  $dx$  to separate variables
- Integrate both sides
- The integral of sine  $x$  is negative cosine  $x$
- The integral of cosine  $x$  is sine  $x$
- Add the constant of integration  $C$

**Answer:**  $y = -\cos(x) + \sin(x) + C$



5. Find the particular solution that passes through the point (0, 1).

$$\frac{dy}{dx} = 3x^2 - 4x$$

- Integrate both sides to find the general solution
- The general solution is  $y$  equals  $x$  cubed minus  $2x$  squared plus  $C$
- Substitute  $x$  equals 0 and  $y$  equals 1
- Solve for  $C$  to get  $C$  equals 1
- Write the particular solution

**Answer:**  $y = x^3 - 2x^2 + 1$

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6. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \frac{1}{x} + e^{2x}$$

- Separate variables by multiplying both sides by  $dx$
- Integrate both sides
- The integral of one over  $x$  is the natural log of absolute value of  $x$
- Use  $u$ -substitution for  $e$  to the  $2x$  to get one-half  $e$  to the  $2x$
- Add the constant of integration  $C$

**Answer:**  $y = \ln|x| + \frac{1}{2}e^{2x} + C$

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7. Find the particular solution that passes through the point (1, 2).

$$\frac{dy}{dx} = 6x^2 + 2$$

- Integrate both sides to find the general solution
- The general solution is  $y$  equals  $2x$  cubed plus  $2x$  plus  $C$
- Substitute  $x$  equals 1 and  $y$  equals 2
- Solve  $2$  equals  $2$  plus  $2$  plus  $C$  to find  $C$  equals negative 2
- Write the particular solution

**Answer:**  $y = 2x^3 + 2x - 2$

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8. Find the general solution to the differential equation.

$$\frac{dy}{dx} = \sqrt{x} + x^2$$

- Rewrite the square root as  $x$  to the one-half power
- Cross-multiply  $dx$  to separate variables
- Integrate both sides using the power rule
- For  $x$  to the one-half, add one to the exponent and divide
- Add the constant of integration  $C$

**Answer:**  $y = \frac{2}{3}x^{\frac{3}{2}} + \frac{x^3}{3} + C$

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9. Find the particular solution that passes through the point (0, 3).

$$\frac{dy}{dx} = e^x + 2x$$

- Integrate both sides to find the general solution
- The general solution is  $y$  equals  $e$  to the  $x$  plus  $x$  squared plus  $C$
- Substitute  $x$  equals 0 and  $y$  equals 3
- Solve 3 equals 1 plus 0 plus  $C$  to find  $C$  equals 2
- Write the particular solution

**Answer:**  $y = e^x + x^2 + 2$

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10. Find the general solution to the differential equation.

$$\frac{dy}{dx} = 5x^4 - 3x^2 + 7$$

- Separate variables by multiplying both sides by  $dx$
- Integrate each term on the right side using the power rule
- Simplify the coefficients
- Add the constant of integration  $C$

**Answer:**  $y = x^5 - x^3 + 7x + C$

